AlphaMark: A Benchmarking Pipeline for Financial Forecast Evaluation

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1 Introduction

The development of predictive signals for financial markets is a central objective in quantitative finance. While the construction of such signals has been extensively studied, their empirical evaluation remains fragmented and often inconsistent across studies. Differences in data preprocessing, portfolio construction, and performance metrics make it difficult to compare results or reproduce findings in a systematic way.

To address this issue, we introduce *AlphaMark*, a benchmarking pipeline for evaluating financial forecasts using profit-and-loss (PnL)–based performance metrics. AlphaMark is designed to take financial time series data and user-defined predictive signals as input and produce a standardized set of evaluation statistics and visual summaries in the form of an automated PDF report.

This document serves as both a technical overview and a user guide for the AlphaMark pipeline. The goal is to enable researchers to easily apply the framework to their own studies, while ensuring consistency, transparency, and comparability across different models, datasets, and research projects in quantitative finance.

2 Assumptions & Scope

AlphaMark is designed as a standardized *benchmarking* system rather than a full trading simulator. The following assumptions define its intended scope of use:

- **Frequency.** The pipeline operates on *daily* data only. Intraday or high-frequency modeling is outside scope.
- **Cross-sectional structure.** Signals are evaluated via daily cross-sectional ranking, not time-series forecasting on individual instruments.
- **Portfolio symmetry.** Portfolios are constructed as long/short with symmetric exposure based on signal sign.
- **Transaction costs.** No transaction costs, slippage, or market impact models are applied unless explicitly injected by the user via targets or bet-size definitions.

- **Execution model.** Trades are assumed to execute at target-defined prices without liquidity constraints.
- **Capital constraints.** No leverage, margin, or funding constraints are modeled.
- **Risk management.** No explicit risk controls (e.g., volatility targeting, sector neutrality, beta neutrality) are enforced unless encoded in the signal or bet-size columns.

These design choices ensure that AlphaMark evaluates *signal quality and forecasting structure*, not execution quality or trading infrastructure. The framework is therefore intended for **research benchmarking and model comparison**, rather than deployable trading strategy validation.

3 AlphaMark Pipeline

3.1 Overview

AlphaMark implements a standardized pipeline for evaluating predictive signals on financial time series using profit-and-loss (PnL)–based metrics. The pipeline operates on *daily* data and follows a fixed sequence: (i) data ingestion, (ii) preprocessing and validation, (iii) signal ranking into quantile portfolios, (iv) daily PnL/statistic computation, (v) aggregation across time, and (vi) automated PDF report generation [1].

3.2 Data Preprocessing

AlphaMark consumes daily cross-sectional datasets containing (at minimum) tickers, one or more signal columns, one or more target forward-return columns, and one or more bet-size columns. Each trading day is processed independently, and all metrics are later aggregated across the evaluation window [1].

Preprocessing includes:

- **Alignment.** Ensure signals, targets, and bet sizes are aligned by date and instrument.
- **Return construction.** Forward log-returns are defined as

$$f_{i,t}^{(h)} = \log\left(\frac{P_{i,t+h}}{P_{i,t}}\right),$$

  where h is the horizon and $P_{i,t}$ is the price used by the target definition.
- **Validation.** Filter days/instruments with missing values or insufficient coverage (e.g., empty-day behavior is handled deterministically by a configured policy).
- **Outlier control.** When enabled, extreme observations in selected metrics are flagged for outlier reporting and robustness checks.

3.3 Signal Processing

A signal $s_{i,t} \in \mathbb{R}$ is a daily forecast for instrument i on day t. AlphaMark uses signals in a cross-sectional manner (ranking within day). To avoid look-ahead bias, signals must be constructed using information available up to day t, while targets are defined on future returns over $[t, t+h]$.

Signals are converted into quantile ranks via a sorting rule. Let $\mathcal{U}_t$ be the instrument universe at day t. For a given quantile level $q \in (0, 1]$, define the selected subset

$$\mathcal{U}_t^{(q)} = \{i \in \mathcal{U}_t : \ |s_{i,t}| \text{ is in the top } 100q\% \text{ of } \{|s_{j,t}|\}_{j \in \mathcal{U}_t}\}.$$

AlphaMark evaluates multiple q values (e.g., $q \in \{1.00, 0.75, 0.50, 0.25\}$), corresponding to quantile buckets in the report [1].

3.4 Portfolio Construction

AlphaMark forms daily long/short portfolios from quantile-ranked signals:

- **Direction.** The position direction for instrument i on day t is $\text{sign}(s_{i,t})$.
- **Bet sizing.** Each position is scaled by a bet size $b_{i,t} \geq 0$ taken from a user-provided bet-size column.

For a chosen target (forward return) $f_{i,t}$ and quantile subset $\mathcal{U}_t^{(q)}$, the daily portfolio PnL is computed as

$$\text{PnL}_t^{(q)} = \sum_{i \in \mathcal{U}_t^{(q)}} \text{sign}(s_{i,t})\, f_{i,t}\, b_{i,t}.$$

AlphaMark also tracks the daily traded notional proxy (used for normalization metrics) as

$$B_t^{(q)} = \sum_{i \in \mathcal{U}_t^{(q)}} b_{i,t}.$$

3.5 Performance Evaluation

AlphaMark evaluates each $(\text{signal}, \text{target}, \text{bet size}, q)$ configuration using the following metrics.

PnL and cumulative PnL. Daily PnL is $\text{PnL}_t^{(q)}$. Cumulative PnL over T days is

$$\text{CumPnL}_T^{(q)} = \sum_{t=1}^{T} \text{PnL}_t^{(q)}.$$

PPD (PnL per dollar traded).

$$\text{PPD}^{(q)} = \frac{\sum_{t=1}^{T} \text{PnL}_t^{(q)}}{\sum_{t=1}^{T} B_t^{(q)}}, \qquad \text{PPD}_{\text{bps}} = 10^4 \cdot \text{PPD}^{(q)}.$$

Sharpe ratio. Let $x_t = \mathrm{PnL}_t^{(q)}$. The annualized Sharpe ratio is

$$\mathrm{SR}^{(q)} = \frac{\overline{x}}{s_x}\sqrt{252},$$

where $\overline{x}$ is the sample mean and s_x is the sample standard deviation.

Hit ratio.

$$\mathrm{Hit}^{(q)} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\{\mathrm{PnL}_t^{(q)} > 0\}.$$

Long ratio.

$$\mathrm{LongRatio}_t^{(q)} = \frac{\sum_{i \in \mathcal{U}_t^{(q)}} \mathbf{1}\{s_{i,t} > 0\}}{|\mathcal{U}_t^{(q)}|}, \qquad \mathrm{LongRatio}^{(q)} = \frac{1}{T}\sum_{t=1}^{T} \mathrm{LongRatio}_t^{(q)}.$$

Size notional.

$$\mathrm{SizeNotional}^{(q)} = \frac{1}{T}\sum_{t=1}^{T} B_t^{(q)}.$$

R^2 and t-stat. A standard diagnostic is the daily cross-sectional regression

$$f_{i,t} = \alpha_t + \beta_t s_{i,t} + \varepsilon_{i,t}.$$

The reported R^2 summarizes explained variation, and the t-stat summarizes the significance of the signal coefficient under the chosen regression/aggregation.

N trades.

$$\mathrm{NTrades}^{(q)} = \sum_{t=1}^{T} |\mathcal{U}_t^{(q)}|.$$

Market correlation. Let m_t be a daily market proxy return (e.g., SPY). Then

$$\mathrm{MarketCorr}^{(q)} = \mathrm{corr}(x_t, m_t), \qquad x_t = \mathrm{PnL}_t^{(q)}.$$

3.6 Visualization and Reporting

AlphaMark generates a PDF report that organizes results into barplots, correlation heatmaps, temporal panels, and (optionally) outlier tables.

Barplot pages. Barplots compare metrics across `target` values, and are faceted by `signal` and `bet_size_col`. Each metric is shown for multiple quantile buckets q (e.g., 100%, 75%, 50%, 25%).

Heatmap 1: signal–signal cross-sectional correlation. For signals a and b, define the daily Spearman correlation across instruments:

$$\rho_t^{\mathrm{XS}}(a,b) = \mathrm{corr}_{\mathrm{Spearman}}\Big(\{s_{i,t}^{(a)}\}_{i\in\mathcal{U}_t}, \{s_{i,t}^{(b)}\}_{i\in\mathcal{U}_t}\Big).$$

The heatmap displays the average $\frac{1}{T}\sum_{t=1}^{T}\rho_t^{\mathrm{XS}}(a,b)$.

H1: smoothed signal–signal correlation over time. Using a rolling window of length W,

$$\overline{\rho}_{t,W}^{\mathrm{XS}}(a,b) = \frac{1}{W}\sum_{u=t-W+1}^{t}\rho_u^{\mathrm{XS}}(a,b).$$

Heatmap 2: time-series Spearman correlation of daily PnL. For a fixed `target`, `bet_size_col`, and quantile bucket q, let $x_t^{(a,q)}$ denote the daily PnL series for signal a. Then

$$\rho^{\mathrm{PnL}}(a,b) = \mathrm{corr}_{\mathrm{Spearman}}\Big(\{x_t^{(a,q)}\}_{t=1}^{T}, \{x_t^{(b,q)}\}_{t=1}^{T}\Big).$$

H2: rolling correlation of daily PnL.

$$\rho_{t,\mathrm{roll}}^{\mathrm{PnL}}(a,b) = \mathrm{corr}_{\mathrm{Spearman}}\Big(\{x_u^{(a,q)}\}_{u=t-W+1}^{t}, \{x_u^{(b,q)}\}_{u=t-W+1}^{t}\Big).$$

Heatmap 3: correlation of daily PnL contribution vectors. Define the daily instrument-level PnL contribution for signal a as

$$c_{i,t}^{(a,q)} = \mathbf{1}\{i\in\mathcal{U}_t^{(q)}\}\,\mathrm{sign}(s_{i,t}^{(a)})\,f_{i,t}\,b_{i,t}.$$

Let $c_t^{(a,q)} = (c_{i,t}^{(a,q)})_{i\in\mathcal{U}_t}$. Heatmap 3 summarizes the correlation of these vectors between signals across time.

H3: expanding correlation of PnL contribution vectors. With an expanding window up to time t,

$$\rho_{t,\mathrm{exp}}^{\mathrm{Vec}}(a,b) = \mathrm{corr}_{\mathrm{Spearman}}\Big(\{c_u^{(a,q)}\}_{u=1}^{t}, \{c_u^{(b,q)}\}_{u=1}^{t}\Big),$$

where the correlation is computed consistently under the implementation's vector-correlation procedure.

Temporal metric pages. For each (`target`, `signal`, `bet_size_col`), AlphaMark generates temporal plots for selected variables (commonly cumulative PnL, cumulative PPD, number of trades, and size notional), with separate lines for each quantile bucket.

Outlier tables. When enabled, AlphaMark produces compact outlier tables for selected metrics (e.g., PnL, PPD, size notional, number of trades), listing the top-k most extreme observations under the chosen ranking rule.

Cross-correlation function (CCF) against market proxy. When enabled, AlphaMark can compute a cross-correlation function between a ticker-level alpha series a_t (raw or PnL-based, depending on configuration) and a market proxy series m_t (e.g., SPY). For lag ℓ,

$$\mathrm{CCF}(\ell) = \mathrm{corr}(a_t, m_{t+\ell}), \qquad \ell \in \{-L, \ldots, L\},$$

where L is the configured maximum lag. This diagnostic is intended to detect lead–lag relationships between alpha behavior and market movements.

Computational note. The CCF computation can be computationally intensive, particularly for long evaluation windows or large maximum lag values. Runtime increases with both the number of time observations and the configured lag range. Even for moderate sample sizes, this diagnostic may take noticeably longer than other metrics. Users are advised to reduce the evaluation interval or lower `ccf_max_lag` when experimenting interactively.

4 Installation and Usage

This section describes how to verify installation, configure the pipeline, and interpret the generated output.

4.1 Input Format and Data Structure

AlphaMark reads daily inputs from Python pickle (`.pkl`) files using a NumPy-version-compatible unpickler. While the loader can unpickle arbitrary Python objects, AlphaMark expects each daily file to unpickle to a tabular object with pandas DataFrame semantics (and in the provided implementation, pandas DataFrames are assumed throughout).

The pipeline reads files from the configured input directory and processes them sequentially by date.

Each daily DataFrame must contain at minimum:

- **Identifier column:** `ticker`
- **At least one signal column**
- **At least one forward-return target column**
- **At least one bet-size column**

The structure is cross-sectional: each row represents one instrument i on day t.

Minimal schema example:

```
ticker      pret_signal     fret_1d     betsize_equal
-------------------------------------------------------
```

```
SPY         0.23            0.0021      1.0
MSFT       -0.11           -0.0014      1.0
GOOG        0.45            0.0033      1.0
AMZN       -0.08           -0.0007      1.0
```

Multiple signals, targets, and bet-size columns are supported within the same file.

By default, the code discovers columns using prefix or regex rules defined in the configuration (e.g., `signal_prefix`, `target_prefix`, `bet_prefix`).

4.2 Test Dataset

A dataset is available for testing and experimentation:

AlphaMark Data Source (Dropbox)

Users may download any number of daily files. Each file corresponds to one trading day.

Place downloaded files into:

```
input/DAILY_FEATURES_PKL/
```

4.3 Project Structure

Expected directory layout:

```
AlphaMark/
  input/
    DAILY_FEATURES_PKL/
  output/
  main.py
  requirements.txt
```

4.4 Environment Setup

Create and activate a virtual environment (recommended):

```
python -m venv venv
source venv/bin/activate        # macOS/Linux
venv\Scripts\activate           # Windows
```

Install dependencies:

```
pip install -r requirements.txt
```

4.5 Configuration

The pipeline is controlled via two dictionaries in `main.py`:

- `DEFAULT_RUNNER_CONFIG`
- `DEFAULT_PLOT_CONFIG`

Key configurable parameters include:

- **Input directory:** `features_input_dir`
- **File pattern:** `features_glob`
- **Signal discovery:** `signal_prefix`, `signal_regex`
- **Target discovery:** `target_prefix`, `target_regex`
- **Bet-size discovery:** `bet_prefix`, `bet_regex`
- **Quantile levels:** `quantiles`
- **Date interval:** `interval_start`, `interval_end`
- **Diagnostics toggles:** `add_spearman`, `ccf_enable`, etc.

Default settings (with the exception of date windows) are sufficient for running the provided dataset. Users should modify quantile levels, diagnostic toggles, and date windows when conducting research experiments.

4.6 Running the Pipeline

From the project root:

```
python main.py
```

Execution performs:

1. Data ingestion and validation
2. Cross-sectional ranking
3. Portfolio construction
4. Daily and summary metric computation
5. Diagnostic generation
6. Automated PDF report creation

4.7 Output

Successful execution generates:

```
output/Quantile_Combined_Report.pdf
```

and associated output files in the `output/` directory.

The PDF report includes:

- Barplots comparing metrics across quantile levels
- Correlation heatmaps between signals
- Time-series plots of cumulative performance
- Optional cross-correlation distributions
- Outlier diagnostics

4.8 Interpreting Results

When evaluating signals, users should consider:

- **Sharpe ratio stability.** Is performance consistent across quantiles?
- **Monotonicity.** Does performance strengthen for higher-magnitude signal subsets?
- **Correlation structure.** Are signals redundant or complementary?
- **Market exposure.** Is PnL correlated with the market proxy?
- **Temporal stability.** Are gains concentrated in specific periods?

Strong signals typically exhibit:

- Positive cumulative PnL
- Sharpe ratios materially above zero
- Performance that increases with signal magnitude
- Limited unintended market correlation

References

[1] M. Cucuringu. *A Pipeline for Benchmarking Forecasts in Financial Time Series Data (PnL Markouts)*. 2025.